35. IoT Communication Protocols & Layers (4M CO3)

Definition:
IoT communication protocols are **a set of rules and standards that allow IoT devices, sensors, and applications to communicate with each other**. These protocols ensure that data collected from sensors reaches the cloud or applications reliably and securely.

IoT Protocol Stack Layers:

Layer	Function	Example
Physical / Perception Layer	Sensors detect real-world data and actuators perform actions	Temperature sensor, motor, light sensor
Link / Data Link Layer	Provides node-to-node communication, error detection, and correction	Zigbee, Bluetooth, LoRa
Network Layer	Provides addressing, routing, and forwarding of data packets	IPv6, 6LoWPAN, RPL
Transport Layer	Ensures end-to-end data delivery and reliability	TCP, UDP
Application Layer	Provides interfaces and communication protocols for applications	MQTT, CoAP, XMPP

Explanation:

- The **Physical Layer** senses the environment.
- The **Link Layer** allows devices to talk to each other in a network.
- The **Network Layer** makes sure data goes to the right place.
- The **Transport Layer** ensures data is not lost.
- The **Application Layer** allows apps to use the IoT data.

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36. Link Layer / Physical Layer Protocols (4M CO3)

Role: This layer deals with **how devices communicate physically and over short ranges**. It handles wireless signals, access control, and error detection.

Protocol	Description	Real-World Example
Zigbee	Low-power, short-range wireless standard	Smart home lighting, security sensors

Protocol	Description	Real-World Example
Bluetooth / BLE	Short-range wireless, low energy	Smartwatches, fitness trackers
6LoWPAN	IPv6 over low-power wireless personal area networks	Environmental monitoring
LoRa / LoRaWAN	Long-range, low-power, suitable for rural IoT	Smart agriculture, water level monitoring

Diagram (Simplified):

Sensor → Zigbee → Gateway → Cloud

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37. Network Layer Protocols (4M CO3)

Role: Responsible for **addressing, routing, and delivery of IoT data** across networks.

Protocol	Description	Example
IPv6 / 6LoWPAN	Assigns addresses for IoT devices; supports large-scale networks	Smart city street sensors
RPL (Routing Protocol for Low-Power and Lossy Networks)	Optimized routing protocol for low-power IoT devices	Industrial IoT sensors
NAT64 / DNS64	Connects IPv6 devices to IPv4 networks	Mixed IoT networks in enterprises

Explanation:

- IoT devices may be **scattered across cities or factories**. Network protocols ensure data reaches the cloud efficiently.

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38. Transport Layer Protocols (4M CO3)

Role: Ensures **reliable end-to-end communication** between devices and applications.

Protocol	Description	Example
TCP	Reliable, connection-oriented protocol; ensures no data is lost	Smart home cameras, live video streaming

Protocol	Description	Example
UDP	Lightweight, connectionless, fast; suitable for low-power devices	Temperature sensors, wearable devices

Explanation:

- TCP is used when reliability is more important than speed.
- UDP is used when speed and low power are more important than reliability.

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39. Application Layer Protocols (4M CO3)

Role: Provides protocols for **IoT applications to communicate and manage data**.

Protocol	Description	Example
MQTT	Publish/subscribe messaging, lightweight	Home automation, industrial IoT dashboards
CoAP	RESTful protocol, works over UDP, for constrained devices	Smart appliances, energy monitoring
XMPP	XML-based, real-time messaging protocol	Chat-enabled IoT devices, remote monitoring

Explanation:

- MQTT is popular because it uses **low bandwidth** and works well for real-time IoT.
- CoAP is suitable for **small devices with limited power and memory**.
- XMPP is used where **instant messaging or alerts** are needed.

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40. How MQTT Facilitates Communication (4M CO3)

Components:

1. **Publisher:** Device that sends data (e.g., temperature sensor).
2. **Broker:** Central server that routes messages to subscribers.
3. **Subscriber:** Device or app that receives messages (e.g., mobile app).

Advantages of MQTT:

- Low bandwidth usage

- Works well on low-power devices
- Supports real-time communication
- Reliable delivery with QoS levels

Diagram:

Temperature Sensor → MQTT Broker → Mobile App

Publisher Broker Subscriber

Explanation:

- MQTT is **perfect for IoT** because sensors often send small data periodically and devices may have **limited battery**.

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41. Difference Between MQTT & HTTP (4M CO3)

Feature	MQTT	HTTP
Communication Model	Publish/Subscribe	Client/Server
Connection Type	Persistent TCP	Request/Response
Bandwidth Usage	Low	High
Real-Time Support	Yes	Limited
Use Case	IoT sensors, automation	Web applications, APIs

Explanation:

- MQTT is designed for IoT and constrained devices.
- HTTP is designed for web applications, not real-time IoT communication.

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42. Protocols in Detail (3M CO3)

1. **MQTT:** Lightweight, publish/subscribe, low bandwidth, ideal for sensors and actuators.
2. **CoAP:** RESTful, uses UDP, for constrained devices with limited power and memory.
3. **XMPP:** XML-based, supports real-time communication, messaging, and presence.

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43. Sensor Network Topologies (5M CO3)

1. Star Topology:

- Central hub controls all devices.
- Simple but single point of failure.
- **Example:** Smart home lighting system.

2. Mesh Topology:

- Each node connects to multiple nodes.
- Provides redundancy and reliability.
- **Example:** Industrial sensor networks, smart factories.

3. Tree Topology:

- Hierarchical parent-child structure.
- Efficient for large-scale IoT networks.
- **Example:** Smart street lighting or city-wide IoT systems.

Diagram:

```
Star:    Hub
       / | \
      N1 N2 N3


Mesh:    N1----N2
          | \ |
         N3----N4


Tree:    Root
       /  \
      N1   N2
     /       \
    N3        N4
```

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44. Security Components for IoT (4M CO3)

1. **Authentication:** Verifies identity of devices or users.
2. **Authorization:** Grants permissions to access specific resources.
3. **Encryption:** Protects data during transmission and storage.
4. **Integrity:** Ensures that data is not altered during communication.
5. **Intrusion Detection:** Monitors and detects unauthorized access or attacks.

Example:

- A smart lock uses authentication to allow only authorized users to open it.

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45. IoT Security Challenges (4M CO3)

1. **Data Privacy:** Sensor data like location or health data may be exposed.
2. **Weak Authentication:** Default passwords or weak credentials make devices vulnerable.
3. **Network Security:** Wireless IoT protocols are prone to hacking and interception.
4. **Scalability:** Managing security for millions of devices is challenging.
5. **Firmware Vulnerabilities:** Outdated software can be exploited by attackers.
6. **Resource Constraints:** Low-power devices cannot run heavy security algorithms.

Explanation:

- IoT devices often operate in **uncontrolled environments**, making them targets for attacks.
- Security needs to balance **performance, cost, and power efficiency**.

✅ **Summary of Unit 3**

- IoT protocols ensure **efficient communication** between devices, gateways, and applications.
- Protocols exist at **all layers: physical, link, network, transport, and application**.
- **MQTT, CoAP, and XMPP** are widely used for IoT communication.
- Sensor networks can use **Star, Mesh, or Tree topologies**.
- IoT security requires **authentication, encryption, and intrusion detection**, but challenges exist due to **data privacy, device limitations, and scalability**.